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Extraction of the energy distribution of traps in organic semiconductor based on diffusion-trapping model under the framework of admittance spectroscopy

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ABSTRACT

Among physical characteristics of organic semiconductors, trap states within their band-gaps need to be understood thoroughly, as they intricately affect device properties and performances. The depth of the trap state energy level significantly changes the curves of the conductance and capacitance versus frequencies in the admittance spectroscopy. A diffusion-trapping admittance model is developed to interpret the capacitance and conductance of organic semiconductor devices at zero bias. And a formula for extracting the energy distribution of the trap state from the conductance versus frequency curve is derived. The accuracy of this method in determining the trap state distribution is verified by numerical simulations. Using our model to fit the experimental data of Cs-doped Al/Cs/MDMO-PPV/ITO devices under zero bias, the results are in relative agreement and the reasonableness of this method is verified.

1. Introduction

Defect levels in the band gap of organic semiconductors are frequently observed to change the slopes of the current-voltage curves and result in an incremental capacitance at low frequencies [1–10]. Detailed information about trap states distributions is required for understanding electronic properties of organic semiconductors and for optimizing performance of organic devices such as organic light-emitting diodes (OLEDs). Therefore, extracting the trap state distribution is of great significance, and a reliable and simple method to determine defect distributions is desirable.

Admittance spectroscopy (AS) has been proven to be a powerful spectroscopic method for characterizing electrical properties of organic materials [6,11]. AS measurements can also be used to determine defect energy distributions [12–15]. As early as 1996, T. Walter et al. had already proposed a method to deduce energy distributions of defects in the band gap of a semiconductor by measuring the complex admittance of the PN junction. They determined the defect distributions of polycrystalline Cu(In,Ga)Se2 thin films by this method [15]. This method is also applied to characterize the deep level of defects states within organic photovoltaics (OPVs) and organic light-emitting diodes (OLEDs) [12,16–18]. J. A. Carr et al. introduced this method into the area of OPVs [13,19], and they characterized the deep defect distribution in poly (3-hexylthiophene) (P3HT) based OPVs [12]. D. C. Tripathi et al. used this approach in deriving the defect distribution of m-MTDATA based organic diodes [17]. This method requires the analysis of Mott–-Schottky plot for obtaining the built-in voltage V_{bi}, and therefore can be carried out when the sample shows a typical Schottky behavior [20]. While the PN junction does not exist in many organic semiconductors. It is therefore not reasonable to derive the trap state information of organic semiconductors by this method in practice. For organic semiconductors, the method for obtaining the trap state distribution is only suitable when it is based on a theoretical model of organic semiconductors.

Following the admittance theory derived by D. Dascălu [2], a formula for determining trap distributions in OLEDs was proposed by T. Okachi et al. The defect distributions in polyfluorene-based Green K light-emitting polymer were extracted using this method [7]. This method of estimating the defective state distribution of OLEDs is not performed at zero bias, but at a relatively high voltage (typically equal to or greater than 10 V) [14,20]. It is to ensure that the device current is a space charge limiting current (SCLC) dominated by the drift current. The current model in the theory of Dascălu et al. only applies to the case of shallow traps. For a shallow trap, the relationship between current density J and voltage V satisfies $J \sim V^2$, the classical Mott-Gerney (MG)

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law. But for the deep trap distribution, the occupancy of the trap level increases and causes distortion of current density–voltage curves [21]. It mainly leads to an abrupt increase of the current slope, $J \sim V^m$ with m > 2 [22]. And that means the method proposed by Okachi et al. will be limited when it is used to extract the distribution of deep energy level traps.

In an attempt to address the restrictions of the two previous methods mentioned above, a method for characterizing trap state distribution at zero bias for organic semiconductors is presented here. When the bias is smaller than V_{bi}, the current through the device is dominated by diffusion current [23]. Therefore, this AS model should consider a combination of carrier diffusion and trapping. In 2008, J. Bisquert et al. analyzed the combination of trapping with diffusion transport in photovoltaic devices [24–27]. Based on the theory of Bisquert et al., this paper deduces a formula to characterize the deep trap state distributions at zero bias. The method proposed here can well reconstruct the trap state distribution of the simulated inputs. Plots of capacitance and conductance versus frequency are obtained by inputting Gaussian-type and exponential-type trap distributions. The experimental data of H. H. P. Gommans et al. [28] are fitted by adjusting the trap parameters and a relatively approximate result is obtained.

2. Theoretical background

2.1. Methods for trap state measurements

The typical AS measurements commonly employed to characterize the energy distribution of the trap states is proposed by Walter et al. [15]. The specific equation for calculating the trap distribution is given by

$$N_t(E_\omega) = -\frac{V_{bi}}{qW}\frac{dC}{d\omega}\frac{\omega}{k_BT} \tag{1}$$

where ω is the angular frequency, q is the elementary charge, C is the capacitance, W is the junction thickness, k_{B} is the Boltzmann's constant, T is the room temperature, E_ω is the demarcation energy and can be calculated with:

$$\omega = \omega_0 = \beta N_c \exp\left(\frac{E_\omega}{-k_BT}\right), E_\omega = k_{\mathrm{B}}T\ln\left(\frac{\omega_0}{\omega}\right) \tag{2}$$

where ω_0 is termed as the attempt-to-escape frequency, N_{c} is the effective density of states in the conduction band, β is the coefficients for electron capture, E_ω is the demarcation energy. The trap states below this energy can contribute to junction capacitance [15,17,19]. The model promoted by Walter et al. is only applicable to samples that exhibit a Schottky behavior, so for many organic semiconductors where PN junctions are not existed, method proposed by Walter et al. may no longer be suitable.

In 2009, Okachi et al. proposed a method for determining trap state distributions in OLEDs by AS on the basis of a theory for single-injection SCLC [7]. The SCLCS impedance formula derived by Dascălu is given by Refs. [1–4]:

$$Z(\omega) = 6\theta\alpha R_i\sum_{k=0}^{\infty}\frac{1}{k+3}\frac{\Gamma(\theta\alpha+1)}{\Gamma(\theta\alpha+k+2)}(-i\omega\alpha\tau)^k \tag{3}$$

where Γ is the Euler-gamma function, $R_{\mathrm{i}} = 4L^3/(9\varepsilon\theta\mu VS)$ is the low-frequency incremental resistance of the diode, S is the active area of the diode. When the trap state is an energy distribution, the trap parameters θ and $\alpha(\omega)$ need to be rewritten as:

$$\theta = \left[1 + \int_{E_v}^{E_c}\frac{\gamma_c(E)}{\gamma_t(E)}dE\right]^{-1} \tag{4}$$

$$\alpha(\omega) = 1 + \int_{E_v}^{E_c}\frac{\gamma_c(E)}{i\omega+\gamma_t(E)}dE \tag{5}$$

where $\gamma_c(E)dE \approx \beta N_t(E)dE$, $\gamma_t(E) = \beta N_c \exp\left[\frac{E_c-E}{-k_BT}\right]$, $N_t(E)$ is the energy distribution of the trap states. At low frequencies the first term in Eq. (3) is dominant, Okachi et al. therefore simplified Eq. (3) and derived the $N_t(E_\omega)$ formula:

$$N_t(E_\omega) = \frac{2R_i\omega}{S_t\upsilon_{\mathrm{th}}\theta k_BT}\frac{\partial}{\partial\omega}\left\{\frac{\omega^2C(\omega)}{[2R_i\omega C(\omega)]^2 + [2R_iG(\omega)-1]^2}\right\} \tag{6}$$

with S_{t} the capture cross section, υ_{th} the electron thermal velocity.

This method promoted by Okachi et al. differs from Eq. (1) in that capacitance and conductance data are measured not at zero bias, but at relatively high operating voltages. For instance, H. Naito et al. demonstrated the determination of the distribution of trap states in the ITO/HTL/Al hole only device in Ref. [20] by Eq. (6). The operating voltage ranges from 20 to 40 V. At this operating voltage, the current of the device is drift current dominated. The current-voltage equation is a slight variation of the MG square law: $J \approx 9\theta\mu\varepsilon V^2/(8L^3)$, with μ the carrier mobility, L the device thickness, ε the dielectric constant. It is applicable for devices with shallow traps, since the trap population is much less than the population of the transport level, and the electric field distribution is not significantly altered [21].

As for the current density affected by deep level traps, the slope of *J-V* is usually greater than 2, which implies that this method might also has restrictions. For organic semiconductors affected by deep energy level trap states, this method may no longer be applicable when the *J-V* does not satisfy the MG law. Moreover, the method of Okachi et al. is generally used to extract defect distributions with exponential or Gaussian tail state shapes, and defect state distributions with complete Gaussian shapes are usually less well obtained. In the next section, the current density model combining the effects of diffusion of carriers and trap states are discussed, as well as the method of extracting trap distribution from it.

2.2. Diffusion and trapping model

The diffusion-trapping ac admittance model developed here is based on the work of Bisquert et al. [24,27]. In order to ignore the case of double carrier recombination, the model discussed here is mainly based on a single carrier injected diode structure. For the purposes of clarification, it is convenient to denote steady-state quantities by $\overline{x}$ and small perturbation quantities by $\widetilde{x}$. The method for measuring the ac electrical response is the ac admittance, which involves an oscillating voltage perturbation of small amplitude $\widetilde{V}$ around a stationary value $\overline{V}$:

$$V = \overline{V} + \widetilde{V}\exp(i\omega t) \tag{7}$$

and the carrier density behaves as:

$$n = \overline{n} + \widetilde{n}\exp(i\omega t) \tag{8}$$

The amplitude of the ac oscillating density is related to the $\widetilde{V}$:

$$\widetilde{n} = \frac{d\overline{n}}{d\overline{V}}\widetilde{V} \tag{9}$$

and this equation can be expressed in terms of the chemical capacitance C_0 of the extended states [29,30]:

$$\widetilde{n_c} = \frac{C_0}{qL}\widetilde{V}, C_0 = \frac{d\overline{Q_c}}{d\overline{V}} = qL\frac{d\overline{n_c}}{d\overline{V}} \tag{10}$$

The ac current $\widetilde{I}$ considering free charge $\widetilde{Q_c}$ and trapped charge $\widetilde{Q_{trap}}$ is given by:

$$\widetilde{I}=\frac{\mathrm{d}\widetilde{Q_c}}{\mathrm{d}t}+\frac{\mathrm{d}\widetilde{Q_{trap}}}{\mathrm{d}t}=i\omega Lq(\widetilde{n}_c+\widetilde{n}_t)\exp(i\omega t) \tag{11}$$

The ac admittance is the ac voltage-to-current ratio, and it has the expression:

$$Y=\frac{\widetilde{I}}{\widetilde{V}}=i\omega Lq\frac{\widetilde{n_c}+\widetilde{n_t}}{\widetilde{V}} \tag{12}$$

The traps do not follow instantaneously the voltage variation, since each trap only communicates with the conduction band. By solving the kinetic equation:

$$\frac{\mathrm{d}n_t}{\mathrm{d}t}=\beta(N_t-n_t)n-\beta n_t N_c\exp\left(\frac{E_c-E}{-k_BT}\right) \tag{13}$$

the relationship between $\widetilde{n_c}$ and $\widetilde{n_t}$ can be obtained likewise Eq. (5):

$$\widetilde{n_t}=\widetilde{n_c}\frac{\beta(N_t-\overline{n}_t)}{i\omega+\beta\left[\overline{n_c}+N_c\exp\left(\frac{E_c-E}{-k_BT}\right)\right]} \tag{14}$$

Combining Eqs. (9), (10) and (14) and substituting into Eq. (12) yields the formula for $Y(\omega)$ in terms of the extended state capacitance:

$$Y(\omega)=\mathrm{i}\omega\left[\frac{\beta(N_t-\overline{n}_t)}{i\omega+\beta\left[\overline{n_c}+N_c\,exp\left(\frac{E_c-E}{-k_BT}\right)\right]}+1\right]C_0=i\omega\left[C_0+C_{\mathrm{traps}}(\omega)\right] \tag{15}$$

The second term $C_{\mathrm{traps}}(\omega)$ is the frequency-dependent capacitance of the trap distribution. If the trap state is a distribution with respect to energy E, assume a distribution of traps $N_t(E)$, with occupancy $f_t(E)$, the number of density of carriers in trap states is

$$n_t=\int_{E_v}^{E_c}N_t(E)f_t(E)dE \tag{15a}$$

The form of the frequency-dependent trap capacitance $C_{\mathrm{traps}}(\omega)$ in this case is given by Bisquert et al.:

$$C_{\mathrm{traps}}(\omega)=\int_{E_v}^{E_c}\frac{C_t(E)}{1+i\omega/\omega_t}dE \tag{16}$$

where $\omega_t=\beta N_c\,exp\left(\frac{E_c-E}{-k_BT}\right)/(1-\overline{f}_t)$, $C_t(\omega)$is given as:

$$C_t(E)=\frac{q^2}{k_BT}N_t(E)\overline{f}_t(1-\overline{f}_t) \tag{17}$$

The steady state occupancy $\overline{f}_t$ can be expressed in terms of the Fermi–Dirac function: $\overline{f}_t=1/\left(1+e^{\frac{E_t-E_f}{k_BT}}\right)$. Substituting Eqs. (16) and (17) into Eq. (15), $Y(\omega)$ can be written as:

$$Y(\omega)=\omega^2\int_{E_v}^{E_c}\frac{C_t(E)\omega_t}{\omega_t{}^2+\omega^2}+i\omega\left[C_0+\int_{E_v}^{E_c}\frac{C_t(E)\omega_t{}^2}{\omega_t{}^2+\omega^2}\right] \tag{18}$$

The real part of $Y(\omega)$ is the conductance $G(\omega)=\omega^2\int_{E_v}^{E_c}\frac{C_t(E)\omega_t}{\omega_t{}^2+\omega^2}$, which contains information about $N_t(E)$. Let $h(\omega,E)=\frac{\omega_t(E)}{\omega_t(E)^2+\omega^2}$, similar to the treatment in Ref. [14], this function can be approximated by a delta function:

$$h(\omega,E)=\frac{k_BT\pi}{2\omega}\delta(E_\omega) \tag{19}$$

$G(\omega)$ can be written as:

$$G(\omega)=\frac{k_BT\pi\omega C_t(E_\omega)}{2}=q^2N_t(E_\omega)\overline{f}_t(1-\overline{f}_t)\frac{\pi\omega}{2} \tag{20}$$

and the trap state distribution $N_t(E_\omega)$ can be written as:

$$N_t(E_\omega)=\frac{2G(\omega)}{\overline{f}_t(1-\overline{f}_t)q^2\pi\omega} \tag{21}$$

It is notable that the E_ω here should be treated similarly to Eq. (2) as:

$$\omega=\omega_t=\beta N_c\exp\left(\frac{E_\omega}{-k_BT}\right)\Big/(1-\overline{f}_t),\ E_\omega=k_BT\ln\left[\frac{\beta N_c}{\omega(1-\overline{f}_t)}\right] \tag{22}$$

Here the admittance is mainly determined by the ac current $\widetilde{I}$ controlled by free and trapped charge density, unlike Walter's theory which needs to account for the existence of PN junctions in inorganic devices. Hence it is more appropriate for the majority of organic semiconductors where PN junctions do not exist. In the next section this paper will select different trap parameters and use Eq. (18) to simulate the *C-f* and *G-f* plots under Gaussian type and exponential type trap state energy distributions. Using Eq. (21) to calculate the reconstructed $N_t(E)$ compared to the input defect distribution, the effectiveness of this method is verified.

3. Discussion

3.1. Numerical simulation and validation

In order to test the plausibility of the trap formula developed in this work, two types of defect distributions are input here for numerical simulation. A Gaussian type of defect:

$$N_t(E)=N_g\exp\left[\frac{(E-E_t)^2}{-2\sigma^2}\right] \tag{23}$$

and an exponential type of defects:

$$N_t(E)=N_e\exp\left[\frac{(E-E_v)}{-k_BT_0}\right] \tag{24}$$

The parameters selected here for numerical calculations are: $\beta=10^{-14}$ cm^2/s, $N_c=10^{20}$ cm^{-3}, Fermi energy level $E_f=0.5$ eV. For the input exponential type of trap, $N_e=10^{36}$ cm^{-3}, the values of trap characteristic temperature T_0 are 600 K, 800 K, 1000 K. For the Gaussian type trap distribution, $N_g=10^{33}$ cm^{-3}. To demonstrate the effect of Gaussian width and trap energy level E_t on the admittance, the variance σ used for the calculations ranges from 0.1 to 0.2 eV and E_t ranges from 0.2 to 0.4 eV.

Fig. (1) shows the *C-f* and *G-f* plots with the effect of exponential type traps for different T_0 inputs. With increasing T_0, the magnitude of either conductance or capacitance will increase. Conductance is a curve that increases exponentially with frequency. And the inset indicates that the capacitance hardly varies with frequency at low frequency range and decreases with frequency at high frequencies.

Fig. (2) illustrates the effect of E_t and Gaussian variance σ on the frequency dependence of capacitance and conductance when the trap is of Gaussian type. The growth of the trap energy level causes the G and C amplitudes to grow by several orders of magnitude, and the capacitance is no longer constant at low frequencies, but there is a significant incremental capacitance. The growth of the Gaussian variance causes a slight increase in the amplitude of capacitance and conductance, but the effect of σ is clearly not as obvious as that of E_t.

To test whether the method of Eq. (21) can extract the defect states from the conductance-frequency plots, the Gaussian and the exponential type of traps are input separately and the reconstructed distributions are obtained for comparison.

From the comparison of the reconstruction results with the input traps in Fig. (3), one can derive that the energy distribution of traps even at deep energy levels, whether of Gaussian or exponential type, can theoretically be reasonably extracted using Eq. (21). This means that the accuracy and applicability of the proposed method in this paper are very good. Comparing with the method of Okachi et al., which can only

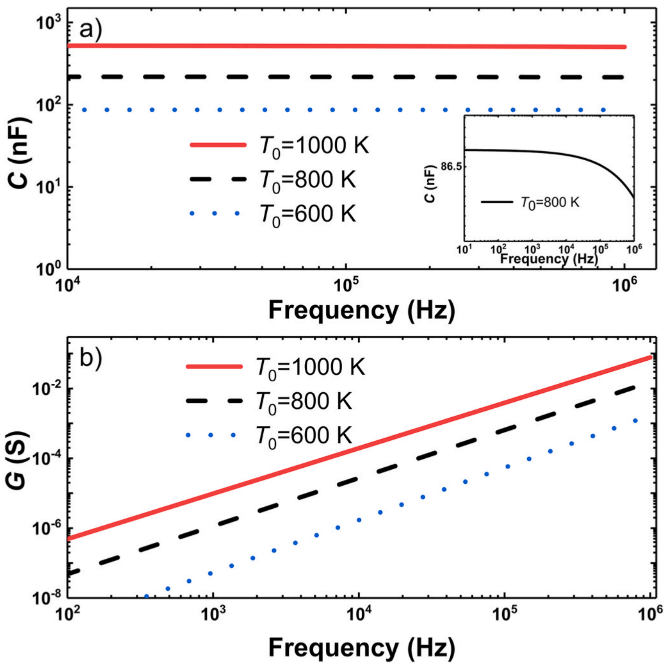

Fig. 1. Frequency dependences of (a) capacitance and (b) conductance calculated using Eq. (18). The inset shows that the capacitance is almost constant with f in the low frequency range and only decreases at high frequencies.

obtain defect state distributions with exponential or Gaussian tail shapes, the method here allows the complete Gaussian shape of the defect state energy distribution to be extracted.

Furthermore, the theory of Okachi et al. is based on the MG law, which requires that the admittance must be measured in the high voltage range and that the current of the device satisfies: $J \approx 9\theta\mu\varepsilon V^2/(8L^3)$. While the admittance model as well as the method mentioned in this paper are discussed at zero bias and will not confront the restriction of current density slopes greater than 2 due to the effects of deep traps. It implies that the applicability of this method is also better.

In the next section this model is used to fit the experimental data at zero bias and to analyze the trap states information.

3.2. *Fitting of experimental data*

In Ref. [28], H. H. P. Gommans et al. measured the capacitance and conductance versus frequency on an Al/Cs/MDMO-PPV/ITO structure for different bias. The film thickness is 170 nm and they measured the frequency-dependent curves of conductance and capacitance at zero bias by varying the deposition coverage of Cs. The experiments were all conducted at a temperature of 20 °C. In this paper, the data of *G-f* and *C-f* experiments are extracted for the Cs deposition coverage of 1.5×10^{14} atoms/cm^2 and 4.8×10^{13} atoms/cm^2, respectively.

When the coverage deposition of Cs is 4.8×10^{13} atoms/cm^2, an exponential type of trap is employed to fit its trap distribution, and the results of the simulation calculated by Eq. (18) are compared with the experimental data. The fitting parameters used here are: $\beta = 10^{-14}$ cm^2/s, $N_c = 3.5 \times 10^{13}$ cm^{-3}, $E_f = 0.5$ eV, $N_t(E) = N_e \exp\left[\frac{(E-E_v)}{-k_B T_0}\right]$, $N_e = 4.5 \times 10^{36}$ cm^{-3}, $k_B T_0 = 0.034$ eV. Fig. 4(a) and (b) demonstrate that the fitted results are in very good agreement.

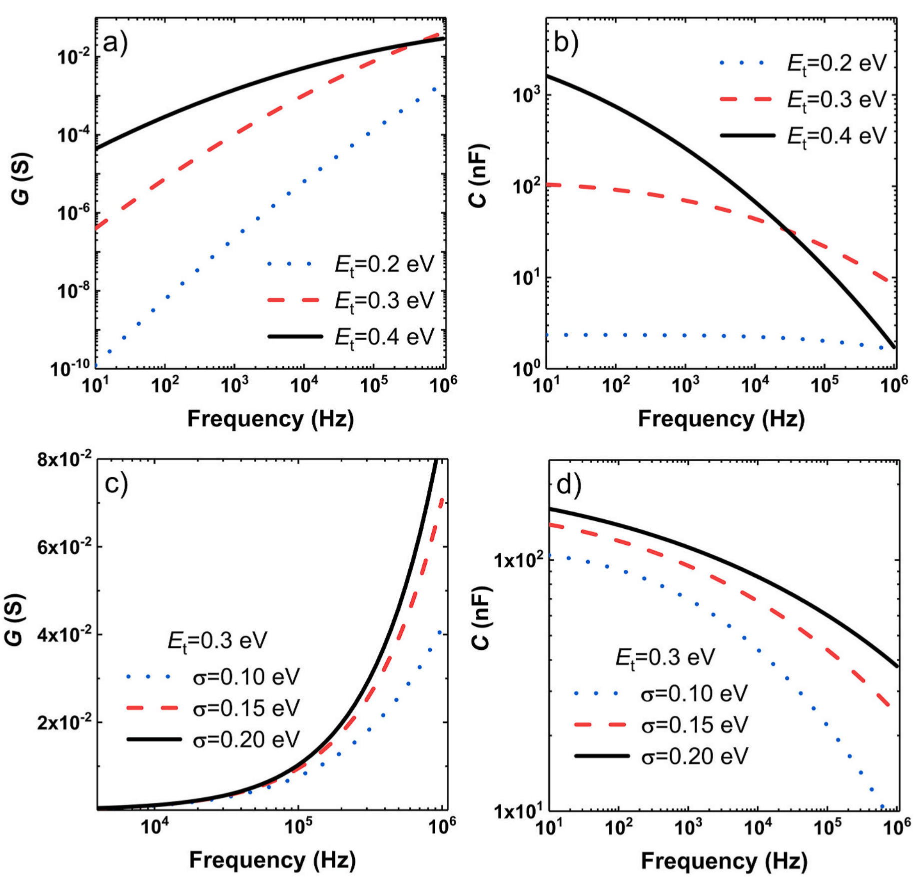

Fig. 2. The calculated *G-f* (a) and *C-f* (b) curves for Gaussian type traps with input variance $\sigma = 0.1$ eV and different E_t. Effect of different Gaussian widths σ on the frequency dependence of conductance (c) and capacitance (d) for the same trap energy level.

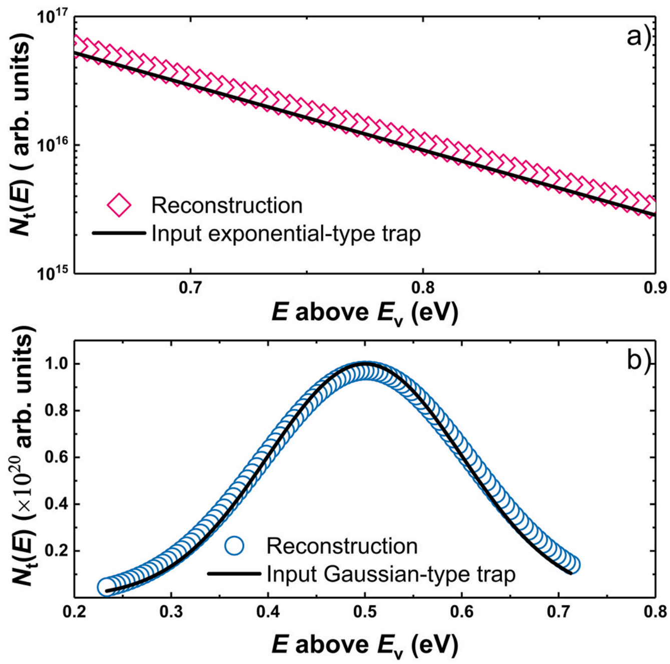

Fig. 3. (a) Comparison of input exponential-type traps with reconstructed results. The trap parameters are: $N_e = 10^{20}$ cm^{-3}, $T_0 = 1000$ K. (b) Comparison of input Gaussian-type traps with reconstructed results. The trap parameters are: $N_g = 10^{20}$ cm^{-3}, trap energy level $E_t = 0.5$ eV, Gaussian width $\sigma = 0.1$ eV.

When the coverage deposition rate of Cs is 1.5×10^{14} atoms/cm^2, the trap distribution is assumed to be a superposition of two Gaussian type distributions. The fitting parameters used here are: $N_t(E) = N_a \exp\left[\frac{(E-E_a)^2}{-2\sigma_1^2}\right] + N_b\, exp\left[\frac{(E-E_b)^2}{-2\sigma_2^2}\right]$, $N_a = 7 \times 10^{31}$ cm^{-3}, $N_b = 1.2 \times 10^{31}$ cm^{-3}, $E_a = 0.325$ eV, $E_b = 0.48$ eV, $2\,\sigma_1{}^2 = 0.0025$ eV, $2\,\sigma_2{}^2 = 0.006$ eV. The results of their fitting are shown in Fig. 4(c) and (d). The simulation results can also be in good agreement with the experimental data. The admittance model demonstrated by Eq. (18) can be used to interpret the trap-affected experimental data at zero bias.

As illustrated in Fig. 5, Eq. (21) is used to extract the trap distributions for the two deposition coverages and compare them to the input simulated trap. One can see that for different Cs deposition coverages, this method can reconstruct the N_t (E) and exhibit the information and structure of trap states distributions.

In order to better use our model to interpret experimental data and extract trap state information, there are some limitations of this model that need to be added. Firstly, the theory of this model is based on single-carrier diode structure to avoid discussing the case of double-carrier recombination. Hence for double-carrier organic semiconductor devices, this model may not be applicable. Secondly, the model presented in the paper neglects the effect of temperature. The experimental data referenced in the manuscript were performed at room temperature. For the case where the experimental temperature deviates far from room temperature, this model may not be suitable and requires deeper investigation. Finally, since it is difficult to accurately obtain the Fermi energy levels of the device in practical experiments, this may influence

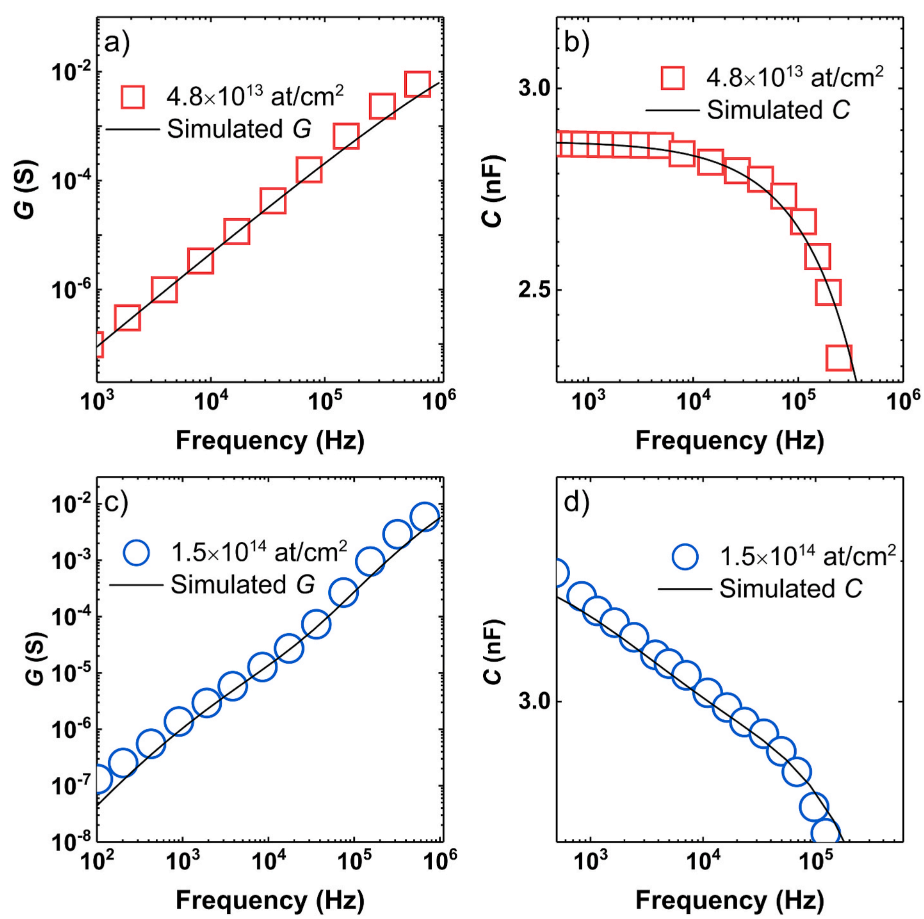

Fig. 4. (a) Comparison of experimental data for Cs deposition coverage of 4.8×10^{13} atoms/cm^2 with simulated *G-f*. (b) Comparison of Cs deposition coverage of 4.8×10^{13} atoms/cm^2 data with simulated *C-f*. (c) Comparison of Cs deposition coverage of 1.5×10^{14} atoms/cm^2 data with simulated *G-f*. (d) Comparison of Cs deposition coverage of 1.5×10^{14} atoms/cm^2 data with simulated *C-f*.

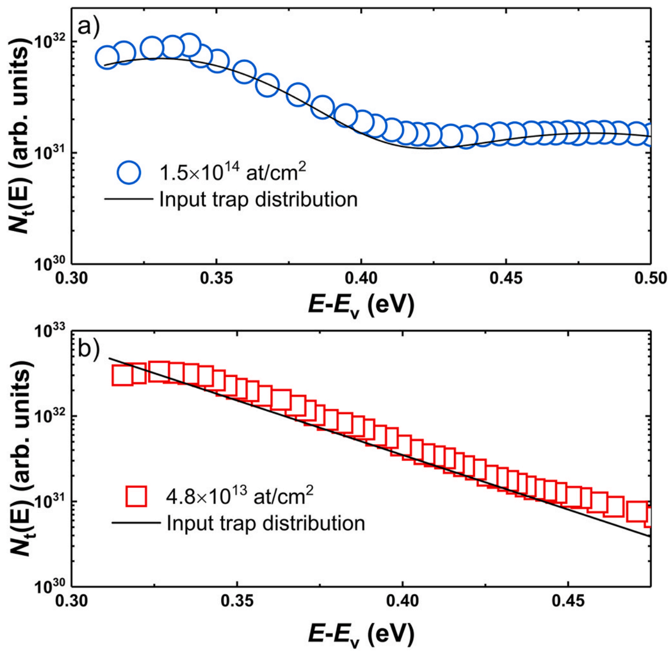

Fig. 5. Input simulated N_t (E) for a deposition coverage of 1.5×10^{14} atoms/cm^2 (a) and a deposition coverage of 4.8×10^{13} atoms/cm^2 (b) versus trap distribution results calculated by Eq. (21).

the accuracy of the calculated N_t (E) amplitude, but will not change the shape of N_t (E). Therefore, in the practical calculation, this possible error could be eliminated by normalizing the calculated N_t (E). The method derived in this paper to characterize the trap energy distribution may be useful in guiding the improvement of the organic semiconductor device preparation process.

4. Conclusion

In conclusion, an admittance model combining diffusion and trap distribution at zero bias is discussed. The effect of Gaussian-type and exponential-type trap energy distributions on the frequency-dependent curves of capacitance and conductance is demonstrated. A formula for extracting trap information from *G-f* curves at zero bias is derived. By adjusting the trap parameters, the fitting results that are relatively consistent with the experimental data are obtained, which verifies the reasonableness of this method.

CRediT authorship contribution statement

Zhendong Ge: Writing – original draft, Data curation. **Tianyou Zhang:** Investigation. **Lei Wang:** Supervision, Funding acquisition.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

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Data availability

The authors do not have permission to share data.

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